A continuous open map of the line
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Theorem 1. *Let $f : \mathbb{R} \to \mathbb{R}$ be continuous, and assume that f is an open mapping. Then f is monotonic: either f is monotone or f is antitone. In fact, the proof shows that f is either strictly increasing or strictly decreasing.*

Proof. Assume that f is continuous and open.

1. The values $f(a)$ and $f(b)$ are distinct for $a < b$. Fix $a < b$, and suppose, toward a contradiction, that $f(a) = f(b)$.

First, no interior point of $[a, b]$ can be a maximum or a minimum of f on $[a, b]$. Indeed, let $c \in (a, b)$. Since (a, b) is open and f is an open map, $f((a, b))$ is an open subset of $\mathbb{R}$ containing $f(c)$. Hence there is $\varepsilon > 0$ such that the ball of radius ε about $f(c)$ is contained in $f((a, b))$.

If $f(x) \le f(c)$ for every $x \in [a, b]$, then $f(c)+\varepsilon/2 \in f((a, b))$, so $f(c)+\varepsilon/2 = f(x)$ for some $x \in (a, b) \subset [a, b]$. This contradicts $f(x) \le f(c)$. Thus c is not a maximizer. Similarly, if $f(c) \le f(x)$ for every $x \in [a, b]$, then $f(c) - \varepsilon/2 = f(x)$ for some $x \in (a, b)$, contradicting $f(c) \le f(x)$. Thus c is not a minimizer either.

Now consider whether there is a point $u \in (a, b)$ with $f(u) \ne f(a)$.

If such a u exists and $f(u) < f(a)$, then, since $[a, b]$ is compact and f is continuous, f attains a minimum on $[a, b]$ at some $d \in [a, b]$. Since $f(d) \le f(u) < f(a) = f(b)$, we cannot have $d = a$ or $d = b$; hence $d \in (a, b)$, contradicting the fact that no interior point is a minimizer.

If instead $f(u) > f(a)$, then f attains a maximum on $[a, b]$ at some $c \in [a, b]$. Since $f(c) \ge f(u) > f(a) = f(b)$, again c is neither a nor b, so $c \in (a, b)$, contradicting the fact that no interior point is a maximizer.

If no such u exists, then $f(x) = f(a)$ for every $x \in (a, b)$. The midpoint $m = (a + b)/2$ lies in (a, b), and $f(x) = f(m)$ for every $x \in [a, b]$: this holds at the endpoints because $f(a) = f(b)$, and in the interior by assumption. Thus m is an interior maximizer, again a contradiction.

Therefore $f(a) = f(b)$ is impossible when $a < b$. It follows that f is injective: if $f(a) = f(b)$ and $a \ne b$, then either $a < b$ or $b < a$, and both possibilities have just been excluded.

2. There are no interior peaks or valleys. Let $a < b < c$. We claim that we cannot have both $f(a) < f(b)$ and $f(c) < f(b)$. Indeed, choose y with

$$\max(f(a), f(c)) < y < f(b).$$

By the intermediate value theorem on $[a, b]$, there is $x_1 \in [a, b]$ with $f(x_1) = y$; by the intermediate value theorem on $[b, c]$, there is $x_2 \in [b, c]$ with $f(x_2) = y$. Since $y < f(b)$, neither x_1 nor x_2 equals b, so $x_1 < b < x_2$. But injectivity gives $x_1 = x_2$, a contradiction.

Similarly, we cannot have both $f(b) < f(a)$ and $f(b) < f(c)$. Choose y with

$$f(b) < y < \min(f(a), f(c)).$$

The intermediate value theorem gives $x_1 \in [a, b]$ and $x_2 \in [b, c]$ with $f(x_1) = f(x_2) = y$. Again $x_1 < b < x_2$, contradicting injectivity.

Consequently, for every $a < b < c$, injectivity and the exclusion of peaks and valleys leave only two possibilities:

$$\big(f(a) < f(b) \text{ and } f(b) < f(c)\big) \quad \text{or} \quad \big(f(c) < f(b) \text{ and } f(b) < f(a)\big). \qquad (*)$$

Indeed, $f(a) \neq f(b)$ and $f(b) \neq f(c)$; of the four sign combinations, the peak and valley combinations are impossible.

3. One increasing pair makes f strictly increasing. Suppose $a < b$ and $f(a) < f(b)$. We prove three auxiliary statements.

First, if $t > a$, then $f(a) < f(t)$. If $a < t < b$, apply $(*)$ to $a < t < b$: the alternative $f(b) < f(t) < f(a)$ is impossible since $f(a) < f(b)$, so $f(a) < f(t) < f(b)$. If $t = b$, this is the assumption. If $t > b$, apply $(*)$ to $a < b < t$: the alternative $f(t) < f(b) < f(a)$ is impossible, so $f(a) < f(b) < f(t)$.

Second, if $t < b$, then $f(t) < f(b)$. If $t < a$, apply $(*)$ to $t < a < b$: the alternative $f(b) < f(a) < f(t)$ is impossible, so $f(t) < f(a) < f(b)$. If $t = a$, this is the assumption. If $a < t < b$, apply $(*)$ to $a < t < b$: the alternative $f(b) < f(t) < f(a)$ is impossible, so $f(a) < f(t) < f(b)$.

Third, if $t < a$, then $f(t) < f(a)$. Apply $(*)$ to $t < a < b$: the alternative $f(b) < f(a) < f(t)$ would contradict $f(a) < f(b)$ in the needed comparison; more precisely, the remaining alternative is $f(t) < f(a) < f(b)$, giving $f(t) < f(a)$.

Now let $x < y$. If $y \leq a$, then $y < b$, and applying $(*)$ to $x < y < b$ gives either $f(x) < f(y) < f(b)$, as desired, or $f(b) < f(y) < f(x)$. The latter is impossible because the second auxiliary statement gives $f(y) < f(b)$. If $a < y$ and $x \leq a$, then $f(x) \leq f(a)$ by the third auxiliary statement (or equality if $x = a$), while $f(a) < f(y)$ by the first; hence $f(x) < f(y)$. If $a < y$ and $a < x$, apply $(*)$ to $a < x < y$: either $f(a) < f(x) < f(y)$, as desired, or $f(y) < f(x) < f(a)$, which contradicts $f(a) < f(x)$ from the first auxiliary statement. Thus f is strictly increasing.

4. One decreasing pair makes f strictly decreasing. Suppose instead that $a < b$ and $f(b) < f(a)$. The argument is symmetric. We prove the corresponding auxiliary statements.

First, if $t > a$, then $f(t) < f(a)$. If $a < t < b$, apply $(*)$ to $a < t < b$: the alternative $f(a) < f(t) < f(b)$ is impossible together with $f(b) < f(a)$, so $f(b) < f(t) < f(a)$. If $t = b$, this is the assumption. If $t > b$, apply $(*)$ to $a < b < t$: the alternative $f(a) < f(b) < f(t)$ is impossible, so $f(t) < f(b) < f(a)$.

Second, if $t < b$, then $f(b) < f(t)$. If $t < a$, apply $(*)$ to $t < a < b$: the alternative $f(t) < f(a) < f(b)$ is impossible, so $f(b) < f(a) < f(t)$. If $t = a$, this is the assumption. If $a < t < b$, apply $(*)$ to $a < t < b$: the alternative $f(a) < f(t) < f(b)$ is impossible, so $f(b) < f(t) < f(a)$.

Third, if $t < a$, then $f(a) < f(t)$. Apply $(*)$ to $t < a < b$: the alternative $f(t) < f(a) < f(b)$ is impossible since $f(b) < f(a)$, so $f(b) < f(a) < f(t)$.

Now let $x < y$. If $y \leq a$, then $y < b$, and applying $(*)$ to $x < y < b$ gives either $f(x) < f(y) < f(b)$ or $f(b) < f(y) < f(x)$. The first is impossible because the second auxiliary statement gives $f(b) < f(y)$, whereas the first would give $f(y) < f(b)$; hence $f(y) < f(x)$. If $a < y$ and $x \leq a$, then $f(y) < f(a)$ by the first auxiliary statement and $f(a) \leq f(x)$ by the third (or equality if $x = a$), so $f(y) < f(x)$. If $a < y$ and $a < x$, apply $(*)$ to $a < x < y$: either $f(a) < f(x) < f(y)$, impossible because the first auxiliary statement gives $f(x) < f(a)$, or $f(y) < f(x) < f(a)$, giving $f(y) < f(x)$. Thus f is strictly decreasing.

5. Conclusion. By injectivity, $f(0) \neq f(1)$. If $f(0) < f(1)$, Step 3 with $a = 0$ and $b = 1$ shows that f is strictly increasing, hence monotone. If $f(1) < f(0)$, Step 4 shows that f is strictly decreasing, hence antitone. Therefore every continuous open mapping of $\mathbb{R}$ into $\mathbb{R}$ is monotonic. $\square$